

AI Engineering Task (Time Limit: < 48 Hrs)

Trademark Class Prediction

Objective

The primary objective of this task is to create an AI model that can accurately predict [trademark classes](#) based on goods and services entered by users. Also refer to the [USPTO ID Manual](#) for the same.

Task

1. Develop an AI model using a technique of your choice to classify trademark classes for goods and services entered by the user. The AI model will be trained using a dataset of existing goods and services from the USPTO ID manual.
 - a. In case you use Transformers, **DO NOT USE** transformers.Trainer API, we want you to write your own training script.
2. Follow good model logging practices, use tools like WandB, MLFlow, TensorBoard etc, and monitor the metrics.
3. Test the AI model with a wide range of inputs, including those that are not present in the ID manual.
4. Make a REST API which can be integrated by other developers to input description of their goods & services and display the recommended classes as a response.
5. Dockerize the application and serve it.
6. The request must have a `user\_id`, if this user is already present in the database, the frequency of the API calls for the user must be incremented by 1. Else, create an entry and it should be assigned as 1.
 - a. In case, this user makes more than 5 requests, it should throw HTTP 429 status code.
7. Also, record inference time for each request, and preferably add API logging for good practices.
8. **Submission:** Submit your final work through GitHub. Add recruitments@trademarkia.com as a contributor on the repository. **Ensure the repository name is `<REGISTER\_NUMBER>\_AI`. All the repositories are to be private.**

Resources

- Watch this [loom](#) for the explanation about this task.

- [Link](#) to dataset.

Instructions:

1. Follow good system design practices
2. GitHub commit history should be clean and representative of the changes in commit.
3. Good documentation should be present.

Constraints:

- **As an intellectual property protection company, we take plagiarism extremely seriously. If your codebase is found to be similar to another submission, or an open source repository, it will be reported to the university.**
- The student is allowed to refer to any and all online resources such as documentation, stackoverflow, etc.
- **The student is required to complete this within 48 hours providing the assignment [the first and last commit need to have been within 48 hours of difference].**
- Allowed Packages: All
 - You are free to use any and all packages but, these are tools we use and hence preferred:
 - API: Django
 - Modeling: PyTorch, Transformers
 - **Note:** In case you use Transformers, **DO NOT USE** transformers.Trainer API, we want you to write your own training script.
- In case you are using some open-source code, you are required to cite the Git repo or paper.